Zijian Liu

+1 (201) 565-5192
zl3067@stern.nyu.edu
zl3067.github.io
Zijian Liu

Education

Sept. 2022 – Jan. 2027 (or May 2027) **Department of Technology, Operations, and Statistics at Stern School of Business, New York University,** NY
Ph.D. Candidate in Operations Management, advisor: Zhengyuan Zhou

Sept. 2019 – May 2021 **Courant Institute of Mathematical Sciences, New York University,** NY
M.S. in Computer Science

Sept. 2015 – June 2019 **School of Computer Science, Wuhan University,** Wuhan, China
B.Eng. in Software Engineering

Experience

May 2026 – Aug. 2026 **Thomas J. Watson Research Center, IBM Research,** NY
Research Internship

Research Interests

Optimization, Sampling, Theoretical Machine Learning

Publications & Manuscripts

(* indicates equal contribution, † indicates alphabetical order)

COLT 2026 **Random Reshuffling Dominates Stochastic Gradient Descent**
Zijian Liu
[arXiv][Proceedings]

ICML 2026 **Can Adaptive Gradient Methods Converge under Heavy-Tailed Noise? A Case Study of AdaGrad**
Zijian Liu
[arXiv][OpenReview]

ICLR 2026 **Clipped Gradient Methods for Nonsmooth Convex Optimization under Heavy-Tailed Noise: A Refined Analysis**
Zijian Liu
[arXiv (Full Version)][OpenReview][Proceedings]

ALT 2026 **Online Convex Optimization with Heavy Tails: Old Algorithms, New Regrets, and Applications**
Zijian Liu
[arXiv (Full Version)][OpenReview][Proceedings]

ICML 2025 **Improved Last-Iterate Convergence of Shuffling Gradient Methods for Nonsmooth Convex Optimization**
Zijian Liu, Zhengyuan Zhou
[arXiv][OpenReview][Proceedings]

ICLR 2025 **Nonconvex Stochastic Optimization under Heavy-Tailed Noises: Optimal Convergence without Gradient Clipping**
Zijian Liu, Zhengyuan Zhou
[arXiv][OpenReview][Proceedings]

ICML 2024 (Oral) **On the Last-Iterate Convergence of Shuffling Gradient Methods**
Zijian Liu, Zhengyuan Zhou
[arXiv][OpenReview][Proceedings]

ICML 2024 **On the Convergence of Projected Bures-Wasserstein Gradient Descent under Euclidean Strong Convexity**
Junyi Fan*, Yuxuan Han*, Zijian Liu, Jian-Feng Cai, Yang Wang, Zhengyuan Zhou
[OpenReview][Proceedings]

ICLR 2024 **Revisiting the Last-Iterate Convergence of Stochastic Gradient Methods**
Zijian Liu, Zhengyuan Zhou
[arXiv (Full Version)][OpenReview][Proceedings]

COLT 2023 **Breaking the Lower Bound with (Little) Structure: Acceleration in Non-Convex Stochastic Optimization with Heavy-Tailed Noise**
Zijian Liu, Jiawei Zhang, Zhengyuan Zhou
[arXiv][Proceedings]

ICML 2023 **High Probability Convergence of Stochastic Gradient Methods**
Zijian Liu*, Ta Duy Nguyen*, Thien Hang Nguyen*, Alina Ene, Huy L. Nguyen
[arXiv (Full Version)][OpenReview][Proceedings]

ICLR 2023 **On the Convergence of AdaGrad(Norm) on $\mathbb{R}^d$: Beyond Convexity, Non-Asymptotic Rate and Acceleration**
Zijian Liu*, Ta Duy Nguyen*, Alina Ene, Huy L. Nguyen
[arXiv][OpenReview]

ICML 2022 **Adaptive Accelerated (Extra-)Gradient Methods with Variance Reduction**
Zijian Liu*, Ta Duy Nguyen*, Alina Ene, Huy L. Nguyen
[arXiv][Proceedings]

ICML 2022 **Distributionally Robust Q-Learning**
Zijian Liu, Qinxun Bai, Jose Blanchet, Perry Dong, Wei Xu, Zhengqing Zhou, Zhengyuan Zhou
[Proceedings]

Operations Research Major Revision **Product Return as a Sequence of Search Processes: Optimality and Search Duration**
Xiaoyu Fan†, Srikanth Jagabathula†, Zijian Liu†, Eitan Muller†
[Available upon Request]

In Submission **Improved Analysis for Stein Variational Gradient Descent**
Zijian Liu, Soumyadip Ghosh, Yingdong Lu, Tomasz Nowicki

In Submission **Adam Converges in Nonsmooth Nonconvex Optimization**
Zijian Liu
[arXiv]

In Submission **In-Expectation Convergence of Stochastic Gradient Methods under Heavy-Tailed Noise**
Zijian Liu
[arXiv]

Preprint **Near-Optimal Non-Convex Stochastic Optimization under Generalized Smoothness**
Zijian Liu, Srikanth Jagabathula, Zhengyuan Zhou
[arXiv]

Preprint **Stochastic Nonsmooth Convex Optimization with Heavy-Tailed Noises: High-Probability Bound, In-Expectation Rate and Initial Distance Adaptation**
Zijian Liu, Zhengyuan Zhou
[arXiv]

Preprint **META-STORM: Generalized Fully-Adaptive Variance Reduced SGD for Unbounded Functions**
Zijian Liu*, Ta Duy Nguyen*, Thien Hang Nguyen*, Alina Ene, Huy L. Nguyen
[arXiv]

Talks

Jul. 2024	*25th International Symposium on Mathematical Programming*
Jul. 2024	*41st International Conference on Machine Learning*
Feb. 2026	*37th International Conference on Algorithmic Learning Theory*
Mar. 2026	*2026 INFORMS Optimization Society Conference*
Apr. 2026	*Thomas J. Watson Research Center, IBM Research*
June 2026	*39th Annual Conference on Learning Theory*

Services

Conference Reviewer
ICML 2022, 2024, 2025, 2026 (Gold Reviewer)
NeurIPS 2024, 2025, 2026
ICLR 2025, 2026
ALT 2025, 2026
SODA 2026

Journal Reviewer
IEEE Journal on Selected Areas in Information Theory (JSAIT)
IEEE Transactions on Automatic Control (TAC)
IEEE Transactions on Control Systems Technology (TCST)
IEEE Transactions on Signal Processing (TSP)
Journal of Machine Learning Research (JMLR)
Management Science
Mathematical Finance
SIAM Journal on Optimization (SIOPT)
Statistica Sinica

Organizer
MOILS seminar (internal seminar series at NYU Stern) Spring 2023

Teaching Experience

(**U** denotes undergraduate-level courses, **G** denotes graduate-level courses)

Summer 2025 — **Operations Management (U),** Instructor — New York University
Class Size: 15
- Design and teach two 3-hour lectures per week independently
- Grade assignments and exams and hold office hours

Fall 2024 — **Operations Management (U),** Teaching Fellow — New York University
Instructor: Prof. Divya Singhvi
- Grade assignments and exams and hold office hours

Summer 2024 — **Stochastic Modeling & Simulation (U),** Teaching Fellow — New York University
Instructor: Prof. Joshua Reed
- Grade assignments and exams and hold office hours

Fall 2020 — **Bayesian Machine Learning (G),** Section Leader — New York University
Instructor: Prof. Andrew Gordon Wilson
- Teach supplementary materials and advanced topics in recitation sessions
- Grade assignments and exams and hold office hours